Data Visualization

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Contents

Web programming: Html, css, and JS

1.1 HTML and CSS

- **SVGS with html**: An SVG (Scalable Vector Graphics) file is an XML-based image format used to display vector graphics. Unlike PNG or JPG images, SVGs scale infinitely without losing quality.

  Key properties:

  - Resolution-independent
  - Small file size for simple graphics
  - Fully stylable with CSS
  - Scriptable with JavaScript
  - Ideal for icons, diagrams, charts, and UI graphics

- **Embedding SVG directly into HTML (inline SVG)**: This is the most powerful and flexible method.

```
<svg width="200" height="100" viewBox="0 0 200 100">
    <rect x="10" y="10" width="180" height="80"
    ↪  fill="steelblue" />
    <circle cx="100" cy="50" r="30" fill="orange" />
</svg>
```

  - <svg> defines the canvas
  - width / height define display size
  - viewBox defines the internal coordinate system
  - Shapes (rect, circle, line, path) are drawn inside

  - Fully stylable with CSS
  - Can be animated
  - JavaScript access to elements
  - Best for interactive graphics

- **SVG elements**:

Element	Purpose
<rect>	Rectangle
<circle>	Circle
<ellipse>	Ellipse
<line>	Line
<polyline>	Connected lines
<polygon>	Closed shape
<path>	Complex shapes
<text>	Text

- **Rect**:
  - **x**: x-coordinate (top-left)
  - **y**: y-coordinate (top-left)
  - **width**: rectangle width
  - **height**: rectangle height
  - **rx**: x-axis corner radius (rounded corners)
  - **ry**: y-axis corner radius
  - **fill**
  - **stroke**
  - **stroke-width**
  - **opacity**
- **Circle**
  - **cx** center x-coordinate
  - **cy** center y-coordinate
  - **r** radius
  - **fill**
  - **stroke**
  - **stroke-width**
- **Ellipse**
  - **cx** center x-coordinate
  - **cy** center y-coordinate
  - **rx** x-radius
  - **ry** y-radius
  - **fill**
  - **stroke**
  - **stroke-width**
- **Line**
  - **x1, y1**: start point
  - **x2, y2**: end point
  - **stroke**: (required)
  - **stroke-width**:
  - **stroke-linecap** (butt, round, square)
  - **stroke-dasharray**:
- **Polyline**
  - **points**: list of coordinate pairs "x1,y1 x2,y2 x3,y3 ..."
  - **fill**: (usually none)
  - **stroke:**
  - **stroke-width:**
  - **stroke-linejoin**:

- **Polygon**
  - **points**: list of coordinate pairs ”x1,y1 x2,y2 x3,y3 ...”
  - **fill**:
  - **stroke**:
  - **stroke-width**:
  - **fill-rule**: (nonzero, evenodd)
- **Text**
  - **x**:
  - **y**:
  - **dx**:
  - **dy**:
  - **text-anchor**: (start, middle, end)
  - **font-family**:
  - **font-size:**
  - **font-weight:**
  - **letter-spacing**:
  - **fill**:
  - **stroke**:
  - **opacity**:
- **SVG viewbox**:

```
<svg viewBox="0 0 200 100">
```

  Means:
  - Coordinate system starts at (0, 0)
  - Width = 200 units
  - Height = 100 units

  This allows scaling without distortion.
- **CSS for <svg>**:
  - fill
  - fill-opacity
  - fill-rule
  - stroke
  - stroke-width
  - stroke-opacity
  - stroke-linecap
  - stroke-linejoin
  - stroke-dasharray
  - stroke-dashoffset

- stroke-miterlimit
- color
- opacity
- x
- y
- cx
- cy
- r
- rx
- ry
- width
- height
- transform
- transform-origin
- transform-box
- font-family
- font-size
- font-style
- font-weight
- letter-spacing
- word-spacing
- text-anchor
- dominant-baseline
- alignment-baseline
- direction
- writing-mode
- display
- visibility
- overflow
- clip-path
- mask
- filter
- cursor
- pointer-events
- animation
- animation-name
- animation-duration
- animation-delay
- animation-iteration-count
- animation-timing-function
- transition

  - transition-property
  - transition-duration

- **Paths**: The <path> element is the most powerful and flexible shape in SVG. Unlike <rect> or <circle>, a path can describe:
  - Straight lines
  - Curves
  - Arcs
  - Complex shapes
  - Icons, symbols, letters
  - Entire illustrations

  It works by following a series of drawing commands stored in the *d* attribute. The *d* string is a mini drawing language, it reads left to right.

  - **Move to (M)**: Moves the "pen" without drawing.

    M x y

    Moves to (x, y)
  - **Line to (L)**: Draws a straight line.

    L x y

    Draws a line from the current point to (x, y)
  - **Close path (Z): Closes the shape by connecting back to the start.**

    Z
  - **Quadratic Curve (Q)**: (cx, cy) = control point (x, y) = end point

    Q cx cy x y

    The control point:
    * pulls the curve
    * bends its direction
    * determines how steep or shallow the curve is

    The curve only passes through:
    * The start point
    * The end point
  - **Cubic Bézier (C)**:

    C x1 y1, x2 y2, x y

    Two control points, more control.
  - **Arc Command (Rounded Shapes) (A)**:

A rx ry x-axis-rotation large-arc sweep x y

Note the difference between lowercase and uppercase control characters

  - **Uppercase**: Absolute position
  - **Lowercase**: Relative movement

- **Triangle with path**:

```
<path d="M 50 10 L 30 80 L 70 80 Z" />
```

- **Fill and stroke**:

```
<path d="M 50 10 L 30 80 L 70 80 Z"
stroke="black"
stroke-width="6"
fill="red"
/>
```

- **Grouping**: The <g> element is used to group multiple SVG elements together so that transformations, styles, or attributes can be applied to them collectively.

```
<svg width="200" height="200">
    <g>
        <!-- SVG elements go here -->
    </g>
</svg>
```

Instead of transforming each element individually, you apply the transformation once to the group.

```
<g transform="translate(50, 50)">
    <circle cx="0" cy="0" r="20" />
    <rect x="30" y="-10" width="40" height="20" />
</g>
```

Both shapes move together. You can apply styles such as fill, stroke, opacity, etc., to all child elements.

```
<g fill="blue" stroke="black" stroke-width="2">
    <circle cx="50" cy="50" r="20" />
    <rect x="90" y="30" width="40" height="40" />
</g>
```

Events applied to a <g> affect all child elements.

```
<g onclick="alert('Clicked group!')">
    <circle cx="50" cy="50" r="20" />
    <rect x="80" y="40" width="30" height="30" />
</g>
```

1.2 Observable

- **Window**: Global variables can be defined with the global window object

```
window.data = ...
```

  Then, this property can be accessed in other cells.

- **Blocks**: In observable, multi-line javascript cells must be placed in a curly brace block
- **HTML templating**: Observable provides an html tag for writing HTML declaratively:

```
html`<h1>Hello</h1>`
```

  - Returns a live DOM node, not a string.
  - HTML is parsed immediately.
  - Values inside ${...} are safely interpolated.

  In a multi statement (curly brace block), this node must be returned to be rendered.

  To make a table dynamically with templating,

```
html`
  <table>
    <thead>
      <tr>
        <th>Year</th>
        <th>Weekday</th>
      </tr>
    </thead>
    <tbody>
      ${data.map(d => html`
        <tr>
          <td>${d.Year}</td>
          <td>${d.Weekday}</td>
        </tr>
      `)}
    </tbody>
  </table>
```

  Notice that html must be placed inside ticks.

1.3 Javascript

- **Var, let, and const**:
  - **Var**: Function-scoped, not block-scoped. Ignores {} blocks such as if, for, and while. Hoisted to the top of the function. Initialized as **undefined**. Can be reassigned ,can be redeclared
  - **Let**: Block-scoped, exists only inside {} where it is defined. Hoisted, but not initialized. Exists in the Temporal Dead Zone (TDZ) until declared. Can be reassigned ,cannot be redeclared in the same scope

```
let x = 3;
x = 4;      // OK
let x = 5; // Error
```

  - **Const**: Block-scoped, same as let. Hoisted but in the TDZ. Must be initialized at declaration

```
const z  = 10; // Good
const y;       // Error
```

    Cannot be reassigned. Const prevents reassignment, not mutation.

- **Immutable types**: These cannot be changed after creation. Any "modification" creates a new value. Primitive Types are all immutable
  - number
  - string
  - boolean
  - null
  - undefined
  - symbol
  - bigint

```
let s = "hello";
s[0] = "H";   // No effect
console.log(s); // "hello"
```

- **Mutable types**: These are objects and collections, whose contents can change without changing the reference.
  - Object
  - Array
  - Function
  - Date
  - Map / Set
- **Pass by value and pass by reference**: JavaScript does not technically have pass-by-reference.

  - Primitive values are passed by value
  - Objects are passed by value of their reference

  All primitives are passed by value.

  Objects are somewhat passed by reference, technically by value of reference. The reference (memory address) is copied, not the object itself.

```
function modify(obj) {
    obj.x = 10;
}

const data = { x: 1 };
modify(data);

console.log(data.x); // 10
```

  - data holds a reference to the object
  - That reference is copied into obj
  - Both point to the same object
  - Mutating the object affects both

- **Objects**: Key value pairs

```
var obj = {x: 2, y: 4}; obj.x = 3; obj.y = 5;
```

  Prototypes for instance functions. We can access properties via dot notation or [] notation. Objects may also contain functions

```
var student = {firstName: "John",
    lastName: "Smith",
    fullName: function() { return this.firstName + " " +
    ↪  this.lastName; }};
student.fullName()
```

  Note: Dot-notation only works with certain identifiers, bracket notation works with more identifiers, like if the key was a string.

- **JSON**: Data interchange format, subset of JS. Uses nested objects and arrays. Data only, no functions.
- **Functional programming in JS**
- **Unary plus operator**: The unary plus (+) operator precedes its operand and evaluates to its operand but attempts to convert it into a number, if it isn't already.
- **Map, filter, and reduce**:
  - **Map**: map transforms each element of an array using a callback function and returns a new array of the same length. It is a pure transformation; the original array is not modified.

```
arr.map((element, index, array) => newElement) -> Array
```

- **Filter**: filter selects a subset of elements based on a predicate function and returns a new array containing only those elements for which the predicate evaluates to true.

```
arr.filter((element, index, array) => boolean) -> Array
```

- **Reduce**: reduce aggregates an array into a single value by repeatedly combining elements using an accumulator function. It is the most general of the three operations.

```
arr.reduce((accumulator, element, index, array) =>
↪  newAccumulator, initialValue) -> any
```

- **forEach**: forEach executes a provided callback function once for each element of an array. It is intended for side effects, not for producing values.

```
arr.forEach((element, index, array) => void) -> void
```

- **Object manipulation**
  - **Deleting properties**: We use the delete operator

```
delete person.age;
```

- **JS functions are objects**: We can assign properties to functions

```
function f() {}

f.x = 10;
f.y = 15;
```

- **Creating html elements manually with JS (SVGs)**:
  - **document.createElementNS**: We use create element to create new elements, to create a new svg element,

```
const new_element = document.createElementNS("http://ww
↪  w.w3.org/2000/svg", "svg");
```

  - **setAttribute**: We can set attributes on our element with setAttribute

```
new_element.setAttribute(name: string, value);
```

  - **appendChild**: We use append child to append child to a root node in our document tree

```
root_element.appendChild(new_element);
```

  - **createTextNode(value)**: Creates a plain text node

```
root_element.appendChild(document.createTextNode("Hello
↪  world"));
```

- **General JS to create HTML dynamically**:
  - **document.createElement(tagName)**: Creates a new DOM element but does not place it in the document.

```
const div = document.createElement("div");
```

  - **document.createTextNode(text)**:

```
const text = document.createTextNode("Hello world");
div.appendChild(text);
```

  - **element.textContent**: Assigns or retrieves text content (preferred in most cases).

```
div.textContent = "Hello world";
```

  - **div.setAttribute("class", "container")**:

```
div.setAttribute("class", "container");
```

  - **getAttribute(name)**:
  - **removeAttribute(name)**:
  - **hasAttribute(name)**:
  - **parent.appendChild(child)**: Appends as the last child.

```
document.body.appendChild(div);
```

  - **parent.insertBefore(newNode, referenceNode)**: Inserts at a specific position.

```
parent.insertBefore(div, parent.firstChild);
```

  - **element.append(...)**:

```
document.body.append(div);
```

  - **element.prepend(...):**
  - **element.before(...):**
  - **element.after(...)**:
  - **element.innerHTML**: Parses an HTML string and replaces the element's contents.

```
div.innerHTML = "<strong>Hello</strong>";
```

  - **document.getElementById(id)**:
  - **document.querySelector(cssSelector)**:

```
const container = document.querySelector(".container");
container.append(div);
```

  - **document.querySelectorAll(cssSelector)**:
  - **element.cloneNode(deep)**: Copies an element.

```
const copy = div.cloneNode(true);
```

  - **element.addEventListener(type, handler)**:

```
button.addEventListener("click", handleClick);
```

- **Direct property assignment (often cleaner and safer):**

```
div.id = "main";
div.className = "container";
```

- **Element properties**:
  - **id**: Unique element identifier
  - **className**: Space-separated CSS classes
  - **classList**: Token-based class API (add, remove, toggle)
  - **tagName**: Uppercase tag name (read-only)
  - **nodeName**: Same as tagName for elements
  - **nodeType**: Numeric node type (1 = Element)
  - **style**: Inline style object
  - **hidden**: Shortcut for display: none

The what why and how

2.1 What

- **Semantics**: Real world meaning of the data
- **Type**: Structural or mathematical interpretation. Both semantics and type often require metadata
  - Item (also Nodes): an entity
  - Link: relationship between two items
  - Attribute: property of an item
  - Position: location in space
  - Grid: how data is sampled
- **Dataset types**:
  - Tables
  - multidimensional data
  - Networks (graphs)
  - Fields (continuous)
  - Geometry (spatial)
  - trees
- **Ordering direction**
  - **Sequential**:
  - **Diverging**: Left and right directions from a common start point
  - **Cyclic**

2.2 Why?

- **Actions**:
  - Analyze: Discover, present, enjoy
  - Produce: Annotate, record, derive
  - Search: Lookup, browse, locate, explore
  - Query: Identify, compare, summarize
- **Targets**:
  - **All data**: Trends, outliers, features
  - **Attributes**:
    - **One**: Distribution, extremes
    - **Many**: Dependency, correlation, similarity
  - **Network data**: Topology, paths
  - **Spatial data**: Shape
- **Visualization for Consumption**: Discover new knowledge, present known information, enjoy. Asking good questions is very important, answers often lead to more questions.
- **Measuring User Experience in Visualization**
  - **Memorability**:
  - **Engagement**:
  - **Enjoyment**:
- **ISOTYPE visualization**: The technique of stacking icons to represent data

2.3 How

- **Marks and channels**:
  - Marks are the basic graphical elements in a visualization
  - Channels are ways to control the appearance of the marks

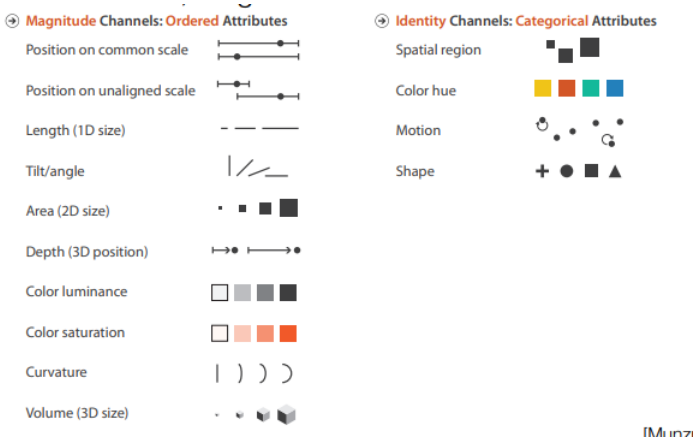
  Marks classified by dimensionality such as points, lines, area, etc. Also can surfaces, volumes.

  Usually map an attribute to a single channel, could use multiple channels but limited number of channels

  Restrictions on size and shape

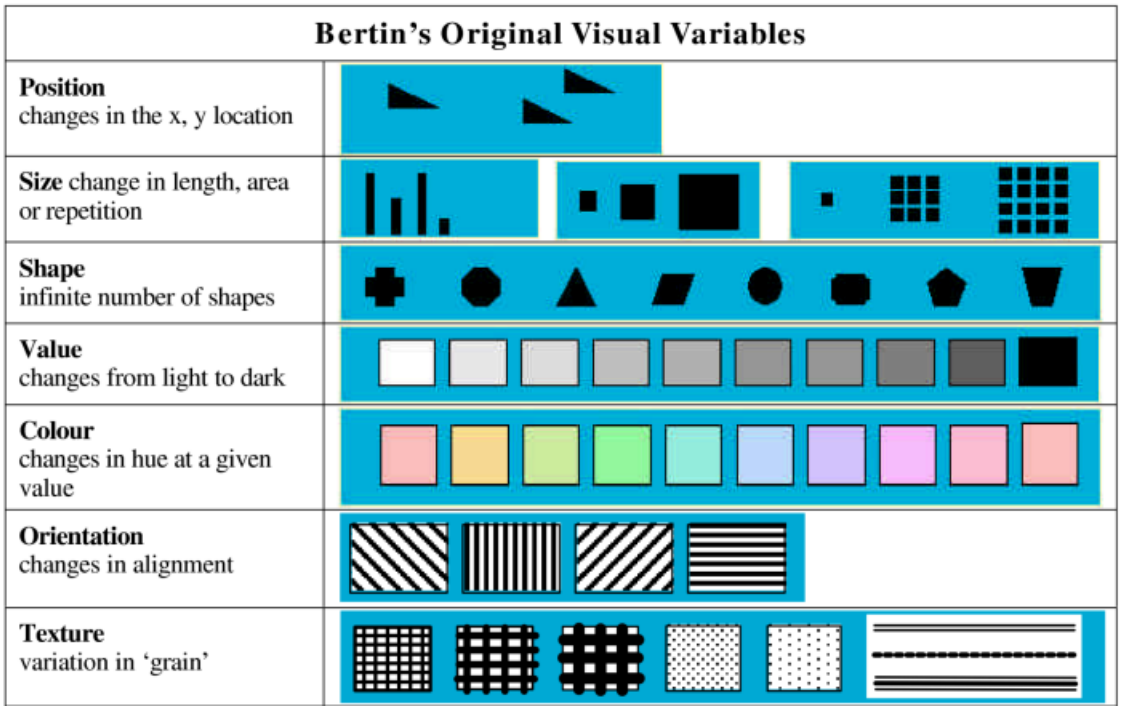
  - Points are nothing but location so size and shape are ok
  - Lines have a length, cannot easily encode attribute as length
  - Maps with boundaries have area, changing size can be problematic

- **Channels**:

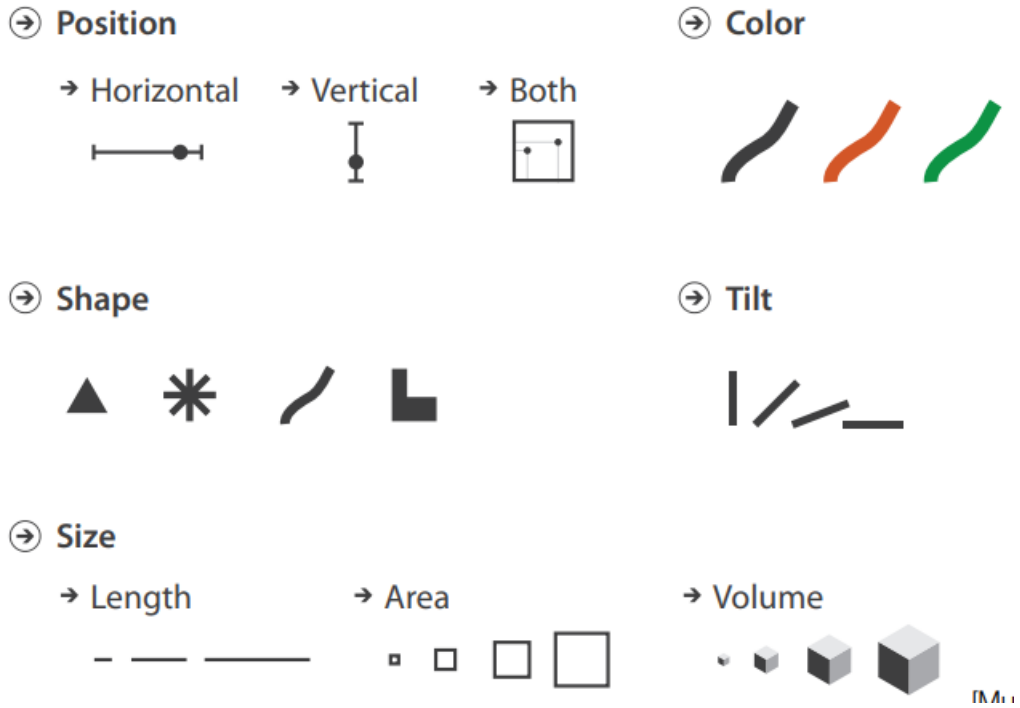


  - **Identity**: What or where
  - **Magnitude**: How much

- **Bertin - Visual Variables**:

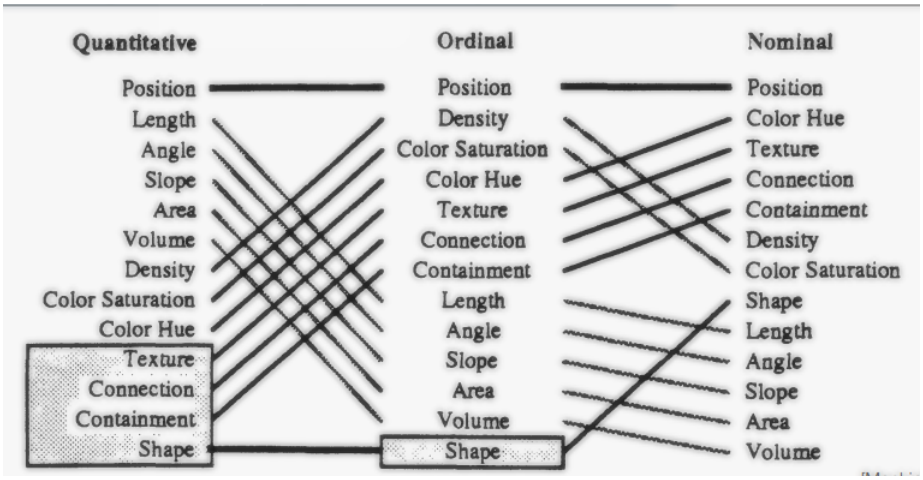


- **More visual channels**:



- **Mark types**: Can have marks for items and links
  - **Connection** $\Longrightarrow$ pairwise relationship

  - **Containment** $\implies$ hierarchical relationship
- **Marks as items / nodes**: Points, lines, areas
- **Marks as links**: Containment, connection
- **Expressiveness and Effectiveness**:
  - **Expressiveness Principle:** all data from the dataset and nothing more should be shown. Don't encode categorical data in a way that implies an ordering
  - **Effectiveness Principle:** the most important attributes should be the most salient
- **Saliency**: How noticeable something is.
- **Mackinlay's Ranking of Perceptual Tasks**:



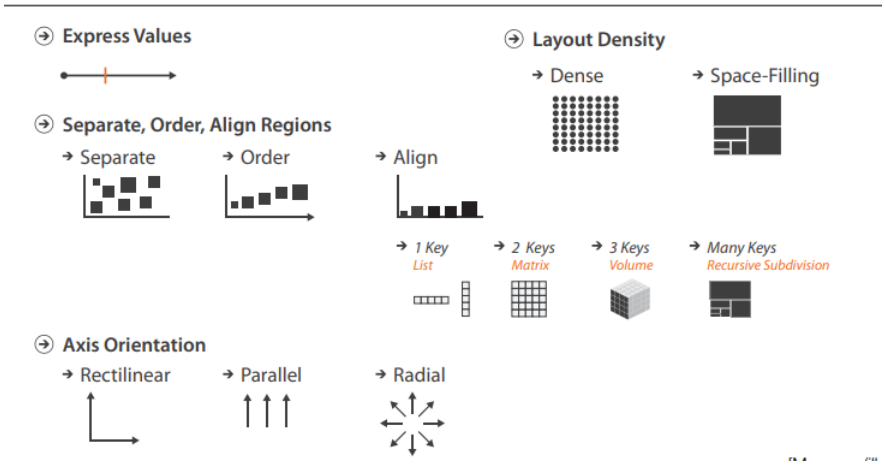
- **Iliinsky's Best Uses, +Ordering, +NumValues**

Example	Encoding	Ordered	Useful values	Quantitative	Ordinal	Categorical	Relational
	position, placement	**yes**	**infinite**	Good	Good	Good	Good
1, 2, 3; A, B, C	text labels	**optional** (alphabetical or numbered)	**infinite**	Good	Good	Good	Good
	length	**yes**	**many**	Good	Good		
	size, area	**yes**	**many**	Good	Good		
	angle	**yes**	medium/few	Good	Good		
	pattern density	**yes**	few	Good	Good		
	weight, boldness	**yes**	few		Good		
	saturation, brightness	**yes**	few		Good		
	color	no	few (< 20)			Good	
	shape, icon	no	medium			Good	
	pattern texture	no	medium			Good	
	enclosure, connection	no	**infinite**			Good	Good
	line pattern	no	few				Good
	line endings	no	few				Good
	line weight	**yes**	few		Good		

- **Discriminability**: Discriminability in data visualization refers to the number of distinct, identifiable steps or levels a visual channel (e.g., color, size, position) can reliably convey to the viewer. It limits how many data categories or magnitudes can be represented before differences become indistinguishable
- **Separability**: Separable means each individual channel can be distinguished
- **Integral**: Means the channels are perceived together
- **Relative vs. Absolute Judgments (Weber's law)**: We judge based on relative, not absolute differences. The amount of perceived difference is relative to the objects magnitude.

2.4 Tabular data

- **Visualization of tables**: Items and attributes. For now, attributes are not known to be positions.
  - **Keys**: Key is an independent attribute that is unique and identifies them Keys are categorical / ordinal
  - **Values**: Tells some aspect of an item. Values are categorical / ordinal / quantitative
  - **Levels**: Unique values of categorical or ordered attributes
- **Arrange Tables**:



- **Express values: Scatterplots**: Two quantitative values, find trends, clusters, or outliers. Marks at spatial position in horizontal and vertical directions.
  - **Correlation**: Dependence between two attributes. Indicated by lines, could be positive or negative.
- **Categorical regions**: Spatial position can be used for categorical attributes. Use **regions**, distinct contiguous bounded areas, to encode categorical attributes.

  Three operations on the regions:

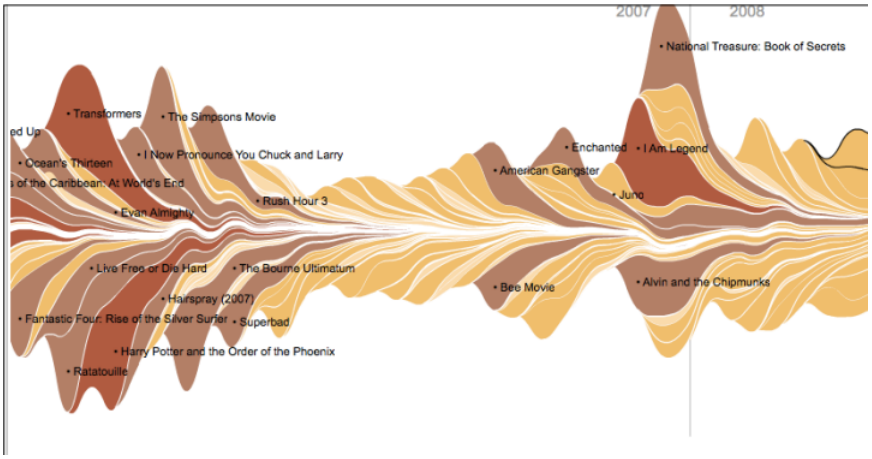
  1. Separate (use categorical attribute)
  2. Align
  3. Order

  Alignment and order can use same or different attribute, some ordered attributes.
- **List alignment; Bar charts**: One quantitative, one categorical. Task is to lookup and compare values. Marks are lines, channels are vertical position (quantitative), and horizontal position (categorical).

The ordering criteria for the x-axis is alphabetical or using quantitative attribute (bar heights sorted). Scalability is hundreds.

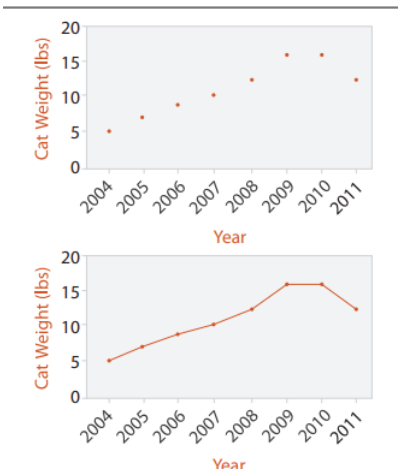
- **Stacked Bar Charts**: Multidimensional table; one quantitative, **two** categorical.
  - **Task**: lookup values, part-to-whole relationship, trends
  - **How**: Line marks, position, subcomponent line marks: length and color.
  - **Scalability**: Main axis (hundreds)
- **Streamgraphs**: Include a time attribute. A streamgraph is a variation of a stacked area chart. Instead of plotting values against a conventional Y axis, the streamgraph offsets the baseline of each "stack" to make it symmetrical around the X axis.
  - **Data:** multidimensional table, one quantitative attribute (count), one ordered key attribute (time), one categorical key attribute

    plus a derived attribute; layer ordering (quantitative)
  - **Task:** analyze trends in time, find (maxmial) outliers
  - **How:** derived position+geometry, length, color
  - **Scalability:** more categories than stacked bar charts



- **Dot and Line Charts**: one quantitative attribute, one ordered attribute
  - **Task**: Lookup values, find outliers and trends
  - **How**: Point mark and positions
  - **Line charts**: Add connection mark (line)

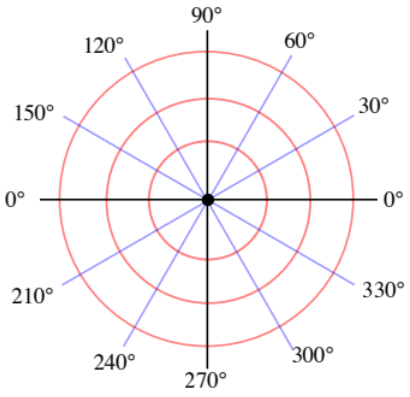
  Similar to scatterplots but allows ordered attributes



Note: Trends in line charts are more apparent because we are using angle as a channel

- **Heatmaps**: Two keys, one quantitative attribute
  - **Task:** Find clusters, outliers, summarize
  - **How:** area marks in grid, color encoding of quantitative attribute

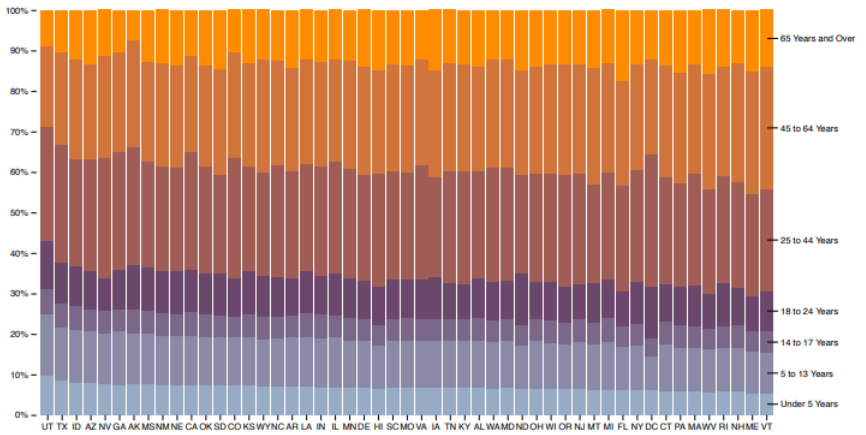
  Red-green color scales often used, be aware of colorblindness
- **Cluster Heatmap**: Data and task same as Heatmap
  - **How**: Area marks but matrix is ordered by cluster hierarchies
- **Dendrogram:** a visual encoding of tree data with leaves aligned
- **Scatterplot Matrix**: A matrix of scatterplots.
  - **Data**: Many quantitative attributes
  - **Derived data**: Names of attributes
- **Radial axis**: Polar coordinates (angle + position along the line at that angle)



  - Radial bar charts
  - Pie charts

  - Donut charts

- **Part-of-whole relative comparisons**:
  - **Normalized stacked bar chart**
  - **Pie chart**:



- **Pie chart vs bar chart**: Angle channel is lower precision then position in bar charts. **We do not read pie charts by angle**, we read pie charts by area, and area is harder to judge than position. Screens are usually not round.

2.5 Color and colormaps

- **Color is a perceptive property:** color depends on the eyes and brain
- **Visible light**: Visible light is a small portion of the electromagnetic spectrum which is composed of waves that at various frequencies (wavelengths), all traveling at the speed of light
- **Important**: We need to be aware of colorblindness when encoding via color

  Our brains may misinterpret color (surrounding colors matter!) even if we aren't colorblind

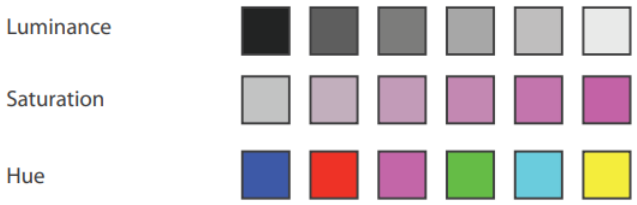
  Be careful! Don't assume that adding color always works the way you intended

  Use known colormaps when possible
- **Color model and color space**: Representation of color using some basis, color space is a collection of color models used for some task.
- **Color gamut**: Subset of colors, defined by corners of color space
- Hue-Saturation-Lightness (HSL) is more intuitive and useful
  - Hue captures pure colors
  - Saturation captures the amount of white mixed with the color
  - Lightness captures the amount of black mixed with a color

  HSL color pickers are often circular
- **Luminance**: It describes the amount of light that passes through, is emitted from, or is reflected from a particular area, and falls within a given solid angle
- **RGB**: Uses three numbers to represent color (red, blue, green)
- **Colormaps**: A colormap specifies a mapping between colors and data values

  Colormap should follow the expressiveness principle

- **Categorical vs. Ordered**: Hue has no implicit ordering, use for categorical data. Saturation and luminance do: Use for ordered data.



- **Categorical Colormap Guidelines**: Don't use too many colors (~ 12), remember your background has a color too. Nameable colors help. Be aware of luminance. Think about other marks you might wish to use in the visualization.

  Often, fewer colors are better. Don't let viewers combine colors because they can't tell the difference. Make the combinations yourself. Also, can use "other" category to reduce the number of colors.

- **Ordered Colormaps**: Used for ordinal or quantitative attributes
  - **Sequential**: $[0, N]$
  - **Diverging**: $[-N, m, N]$, where m is the midpoint, usually 0. Midpoint should be meaningful.

  Can use hue, saturation, and luminance. Can be **continuous (smooth)**, or **segmented (sharp boundaries)**. Segmented matches with ordinal attributes, can be used with quantitative data too.

- **Types of Tasks**: Locate/Explore Identify: Highest Point (Global, In Region), 275m
  - **Locate/Explore and Compare:** Height Compare/Rank
  - **Explore and Identify:** Steepest
  - **Lookup and Identify:** Lookup
  - **Explore and Compare:** Steepness Compare/Rank
  - **Browse and Summarize:** Average Height
  - **Browse and Compare:** Compare Average Height
  - **Combination:** Steepest at 355m

- **Note**: Don't Use Rainbow Colormaps, other colormaps work better. Rainbow colormaps (often called "jet" colormaps) are widely discouraged in data visualization because they introduce perceptual distortions that can mislead interpretation. The problem is not aesthetic; it is rooted in human visual perception and color science.

  A good colormap should change perceptually uniformly with the data. That is, equal changes in data values should appear as equal visual changes.

  Rainbow colormaps violate this property.
  - Some regions (e.g., yellow–green) appear much brighter than others.
  - Other transitions (e.g., blue $\rightarrow$ cyan) appear subtle even when the data difference is large.

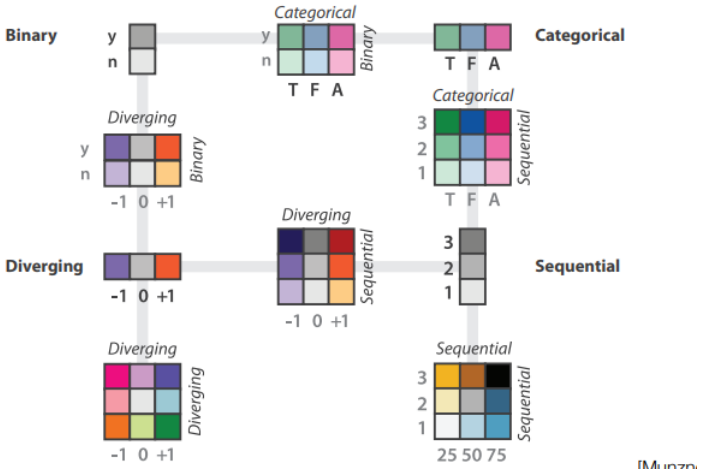
  Small differences in certain ranges appear exaggerated. Large differences in other ranges appear suppressed.

Also, rainbow maps introduce false gradients that do not exist in the data.

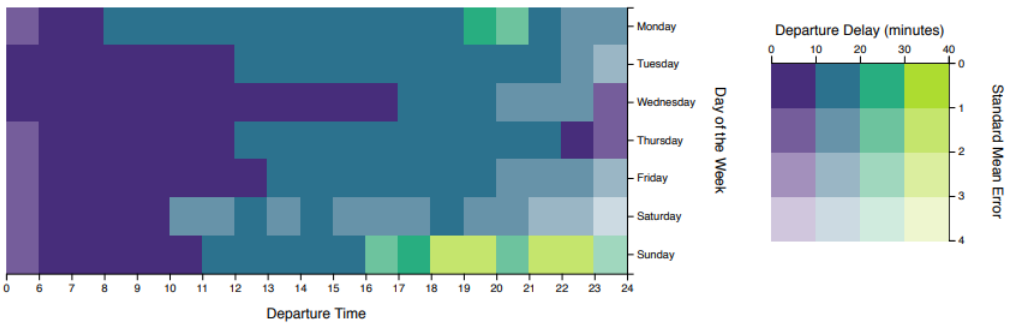
  - **Issues:**
    - Order is unclear
    - Luminance issues
    - Artifacts from discontinuities in luminance
  - Attempts to fix:
    - Isoluminant rainbow
    - Turbo

- **Main ideas**:
  1. **Don't dance around the color wheel**
  2. **Warm colors and blue**
  3. **Avoid too little contrast to background**

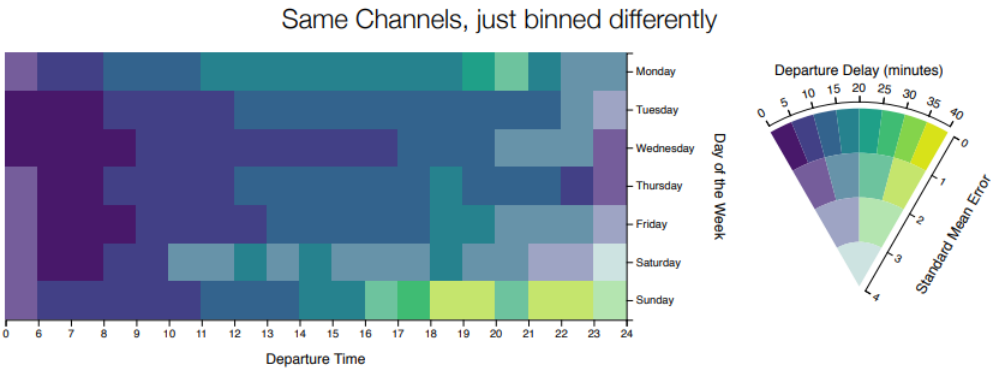
- **Bivariate Colormaps**:



- **Uncertaintly can map to saturation (bivariate colormap)**



- **Value-Suppressing Uncertainty Palette (VSUP)**: Same channels, binned differently.

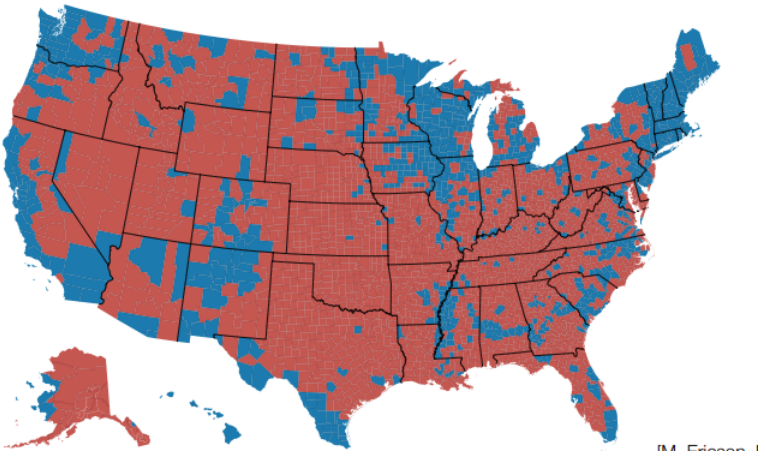


– Legend shape has no significant effect
– Some indication that people avoid high uncertainty with VSUPs
– Tradeoff is that people do choose targets with higher danger when using a VSUP
– VSUPs present uncertainty information simultaneously (superimposed) instead of juxtaposed
– VSUPs encode value and uncertainty via discrete, quantized bins instead of continuously

2.6 Geospatial data

- **Spatial data**: Has positions, may also have non-spatial attributes associated with items
- **Cartography**: Science of drawing maps
- **Goals**: Respect cartographic principles, understand data with geographic references with the visualization principles
- **Adding Data**:
  - **Discrete**: A value is associated with a specific position
    - Size
    - Color hue
    - Charts
  - **Continuous**: Each spatial position has a value (fields)
    - Heatmap
    - Isolines
- **Isolines**:
  - Scalar fields:
    - value at each location
    - sampled on grids

  Isolines use derived data from the scalar field, interpret field as representing continuous values, derived data is geometry: new lines that represent the same attribute value
- **Choropleth (Two Hues)**:



  - **Data:** geographic geometry data one quantitative attribute per region
  - **Tasks:** trends, patterns, comparisons
  - **How:** area marks from given geometry, color hue/saturation/luminance
  - **Scalability:** thousands of regions

D3

- **Data-driven documents**: D3.js (Data-Driven Documents) is a JavaScript library for producing dynamic, interactive data visualizations in web browsers using standard web technologies — primarily HTML, SVG, and CSS. Unlike charting libraries that provide predefined plots, D3 is a low-level visualization toolkit that gives fine-grained control over how data maps to visual elements.

  D3's central abstraction is that data drives document structure. You associate data with elements on a page, and D3 computes what should be created, updated, or removed.

  Conceptually,

  $$\text{Data} \rightarrow \text{Visual elements}$$

- **Key features**:
  - Supports data as a core piece of Web elements
    - Loading data
    - Dealing with changing data (joins, enter/update/exit)
    - Correspondence between data and DOM elements
  - Selections (similar to CSS) that allow greater manipulation
  - Method chaining
  - Integrated layout algorithms
  - Focus on interaction support
- **Creating and returning svg in observable**

```
{
    let svg = d3.create("svg")

    return svg.node();
}
```

- **Data Binding (Selections)**: D3 can bind arrays or objects directly to DOM elements.

```
d3.selectAll("p")
.data([10, 20, 30])
.text(d => d);
```

  This replaces each paragraph's text with the corresponding data value.

- **Select and selectAll**:
  - **d3.select(selector)**: Selects the first element that matches the specified selector string. If no elements match the selector, returns an empty selection. If multiple elements match the selector, only the first matching element (in document order) will be selected.

If the selector is not a string, instead selects the specified node; this is useful if you already have a reference to a node

- **selection.select(selector)**: For each selected element, selects the first descendant element that matches the specified selector string

  If the selector is a function, it is evaluated for each selected element, in order, being passed the current datum (d), the current index (i), and the current group (nodes),

- **d3.selectAll(selector)**: Selects all elements that match the specified selector string. The elements will be selected in document order (top-to-bottom). If no elements in the document match the selector, or if the selector is null or undefined, returns an empty selection.

  If the selector is not a string, instead selects the specified array of nodes

- **selection.selectAll(selector)**: For each selected element, selects the descendant elements that match the specified selector string. The elements in the returned selection are grouped by their corresponding parent node in this selection.

- **selection.selectAll with function example**

```
function run5b(svg) {
    // arrow functions
    svg.selectAll("rect")
        .attr("x", 0)
        .attr("y", (d,i) => i*90+50)
        .attr("width", (d,i) => i*150+100)
        .attr("height", 20)
        .style("fill", "steelblue")
}
```

- **Enter-update-exit pattern**: When data size changes, D3 determines
  - **Enter:** New data → create elements
  - **Update:** Existing data → modify elements
  - **Exit:** Removed data → delete elements

- **Changing attributes**:

```
var r1 = svg.select("rect").attr("width",
↪  100).attr("height", 100);
```

- **Changing styles**:

```
var r1 = svg.select("rect").style("fill", "blue");
```

- **Binding data**: We use

```
selection.data(data: arr, key?: fn)
```

Binds the specified array of data with the selected elements, returning a new selection that represents the update selection: the elements successfully bound to data. Also defines the enter and exit selections on the returned selection, which can be used to add or remove elements to correspond to the new data.

the key parameter in .data(data, key) specifies how data items are matched to existing DOM elements during a data join.

If we write

```
selection.data(data)
```

D3 matches data to elements by index (position):

- First data item → first element
- Second → second

etc... With a key,

```
selection.data(data, key)
```

D3 matches elements by identity, not position. The key function returns a unique identifier for each datum.

```
(d, i) => keyValue
```

Suppose the data is

```
const data = [
    { id: 101, name: "Alice" },
    { id: 102, name: "Bob" }
];
```

Then, the key could be

```
svg.selectAll("circle")
.data(data, d => d.id);
```

D3 now tracks circles by id, not position.

- **The enter selection**: In D3, the enter selection represents data items that do not yet have corresponding DOM elements. It is the mechanism by which new visual elements are created when the dataset grows.

After a data join,

```
const sel = selction.data(data);
```

D3 partitions the result into three disjoint subsets

- Update — data matched to existing elements
- Enter — data with no element yet
- Exit — elements with no data anymore

.enter() returns a special selection containing placeholder nodes — not actual DOM elements. Each placeholder corresponds to one new data item. You cannot style or position them until you append real elements.

```
selection.data(data)
    .enter()
    .append("circle");
```

“For each new datum, create a new <circle> element.”

```
function run8(svg) {
    var selection = svg.selectAll("rect")
    .data([127, 61, 256, 71])

    selection
        .attr("x", 0)
        .attr("y", (d,i) => i*90+50)
        .attr("width", d => d)
        .attr("height", 20)
        .style("fill", "steelblue")

    selection.enter().append("rect")
        .attr("x", 10) // let's just put it somewhere
        .attr("y", 10)
        .attr("width", 30)
        .attr("height", 30)
        .style("fill", "green")
}
```

- **Appending**:

```
selection.append(type)
```

If the specified type is a string, appends a new element of this type (tag name) as the last child of each selected element, or before the next following sibling in the update selection if this is an enter selection. The latter behavior for enter selections allows you to insert elements into the DOM in an order consistent with the new bound data

If the specified type is a function, it is evaluated for each selected element, in order, being passed the current datum (d), the current index (i), and the current group (nodes), with *this* as the current DOM element (nodes[i]). This function should return an element to be appended. (The function typically creates a new element, but it may instead return an existing element.)

- **Inserting**:

```
selection.insert(type, before?)
```

If the specified type is a string, inserts a new element of this type (tag name) before the first element matching the specified before selector for each selected element. For example, a before selector :first-child will prepend nodes before the first child. If before is not specified, it defaults to null.

- **Merge**: .merge() combines two selections into one — most commonly the enter selection and the update selection — so that you can apply the same operations to both newly created and already existing elements.

```
function run10(svg) {
    var selection = svg.selectAll("rect")
    .data([127, 61, 256, 71]);

    selection.enter().append("rect")
        .merge(selection)
        .attr("x", 0)
        .attr("y", (d,i) => i*90+50)
        .attr("width", d => d)
        .attr("height", 20)
        .style("fill", "steelblue");
}
```

Here, the enter section is merged with the update section.

- **Join**: .join() is a high-level method that performs the complete data join lifecycle — handling enter, update, and exit selections in a single operation. It is the modern replacement for the classic pattern using .enter(), .merge(), and .exit().

  Formally, .join() orchestrates:

  - Creation of elements for new data (enter)
  - Updating of existing elements (update)
  - Removal of obsolete elements (exit)

```
selection.data(data).join("tag")
```

For example,

```
svg.selectAll("circle")
    .data(data)
    .join("circle");
```

For each data item, ensure there is exactly one <circle> element.

Internally:

- Enter → append <circle>
- Update → keep existing circles
- Exit → remove extra circles

After .join(), you typically chain attribute setters:

```
svg.selectAll("circle")
    .data(data)
    .join("circle")
    .attr("r", d => d);
```

The selector in selectAll() determines which existing elements participate in the join; the tag in .join() determines what new elements are created.

This affects both newly created and existing elements. This simplifies the classic no-join pattern

```
const update = svg.selectAll("circle") .data(data);

update.enter()
    .append("circle")
    .merge(update)
    .attr("r", d => d);

update.exit().remove();
```

For

```
svg.selectAll("rect")
    .data(data)
    .join("circle");
```

- **Update selection**: Existing <rect> elements are matched to data

They remain <rect> elements (they are not converted)

- **Enter selection**: For new data items, D3 creates <circle> elements
- **Exit selection**: Extra <rect> elements (without data) are removed

- **.join with function parameters**: .join also has two optional parameters `update` and `exit`, specifying functions to call on update and exit. Note that the enter parameter can also be a function, and as is called for enter.

```
svg.selectAll("rect")
.data([127, 61, 256, 71])
.join(enter => enter.append("rect")
       .attr("x", 200)
       .attr("y", 200)
       .attr("width", 10)
       .attr("height", 10)
       .style("fill", "red")
       .call(update_bars),
     update => update.call(update_bars),
     exit => exit.attr("opacity", 1)
       .transition()
       .duration(3000)
       .attr("opacity", 0)
       .remove());
```

- **Remove**:

```
selection.remove()
```

Removes the selected elements from the document. Returns this selection (the removed elements) which are now detached from the DOM. There is not currently a dedicated API to add removed elements back to the document; however, you can pass a function to selection.append or selection.insert to re-add elements.

- **Exit**: The .exit() selection represents DOM elements that no longer have corresponding data after a data join.

```
sel.exit()
```

returns a selection of actual DOM nodes (unlike enter placeholders) that are now obsolete. These nodes still exist in the DOM until you remove them.

```
sel.exit().remove();
```

Is most commonly used, removes all elements that no longer represent any data.

- **Transition**:

```
selection.transition(name?)
```

Returns a new transition on the given selection with the specified name. If a name is not specified, null is used. The new transition is only exclusive with other transitions of the same name.

selection.transition() initiates an animated transition on the selected elements, allowing their attributes, styles, or properties to change smoothly over time rather than instantaneously.

```
d3.select("circle")
    .transition()
    .attr("r", 50);
```

Calling .transition():

1. Captures current state of each element
2. Schedules animation frames
3. Interpolates values over time
4. Applies intermediate values continuously

If nothing is specified,

- **Duration:** 250 ms (approximate default)
- **Delay:** 0
- **Easing:** cubic in-out

We have the functions

```
.duration(ms), .delay(ms), .ease(fn)
```

.ease() takes an ease function, for example d3.easeSin().

- **What can be animated**: Transitions interpolate numeric or color values. Common targets are
  - SVG attributes (cx, r, x, y, etc.)
  - CSS styles (opacity, fill, etc.)
  - Text content (with special handling)
  - Transforms
- **Chaining Transitions**: You can sequence animations

```
.transition()
.duration(500)
.attr("r", 20)
.transition()
.duration(500)
.attr("r", 5);
```

Second transition begins after the first completes.

```
function run15(svg) {
    var selection = svg.selectAll("rect")
        .data([127, 61,256,128])

    selection.enter().append("rect")
            .attr("x", 200)
            .attr("y", 200)
            .attr("width", 10)
            .attr("height", 10)
            .style("fill", "red")
        .merge(selection)
        .transition()
        .duration(3000)
            .attr("x", 0)
            .attr("y", (d,i) => i*10+50)
            .attr("width", d => d)
            .attr("height", 8)
            .style("fill", "steelblue")
            .transition()
            .duration(3000)
                .style("fill", "green")
                .attr("width", d => d*1.5)

    selection.exit()
        .attr("opacity", 1)
        .transition()
        .duration(3000)
            .attr("opacity", 0)
            .remove()
}
```

- **.call()**: .call() invokes a function on a selection (or transition) and passes that selection as the function's argument.

```
selection.call(function, ...args?)
```

```
selection.call(f);
```

Is essentially the same as

```
f(selection)
```

but .call() returns the original selection, enabling chaining.

- **Data as the key**: Consider

```
.data(data, d => d);
```

So, the key is simply the datum value itself. The key function computes a unique key for each datum. It is appropriate when your data consists of primitive unique values, such as numbers or strings.

Consider the initial data

```
[1,2,3]
```

Then, we update the data to

```
[3,2,1]
```

Without the key, D3 assumes

- Element 1 now represents 3
- Element 2 still represents 2
- Element 3 now represents 1

With $d \implies d$,

- Element with key 3 already exists $\rightarrow$ move/update
- Element with key 2 $\rightarrow$ unchanged
- Element with key 1 $\rightarrow$ move/update

- **Nested selections, grouping**:

```
function run19(svg) {
  var myData = [
    [15, 20],
    [40, 10],
    [30, 27]
  ]

  // First selection (within svg)
  var selA = svg.selectAll("g")
    .data(myData)
    .join("g")
    // .attr("y", (d,i) => i*100 + 50)
    .attr("transform", (d, i) => `translate(70,${i*100+50})`)
    // backticks (``) denote template literal
    // normal strings except that text inside ...
    // is evaluated (can be js expression)!

  // Second selection (within first selection)
  var selB = selA.selectAll('circle')
    .data(d => d)
    .join("circle")
    .attr("cx", (d, i) => i*80)
    .attr("r", (d,i) => d)
};
```

- **Scaling**: In D3, scaling refers to mapping values from a data domain to a visual range — typically pixel positions, lengths, colors, or sizes. Raw data values usually do not correspond directly to screen coordinates.

  Every D3 scale has two fundamental parts:

  1. **Domain**:

     ```
     [min data, max data]
     ```

  2. **Range**

     ```
     [min pixel, max pixel]
     ```

  For example,

  ```
  const x = d3.scaleLinear().domain([0,100]).range([0,500]);
  ```

  A scale is simply a function. Now, we can use it when defining x, y attributes.

  ```
  .attr("x", d => x(d));
  ```

- **scaleLog and scaleBand**: A log scale maps data using a logarithmic function, instead of using a linear mapping

$$y = ax + b,$$

it uses a logarithmic one

$$y = a \log x + b.$$

We use logarithmic scaling when data spans multiple orders of magnitude. A linear scale would compress small values and stretch large ones excessively.

Note: Be careful with the domain,

$$D(\log(x)) = (0, \infty).$$

We can also define the base of the logarithm

```
x.base(2);
```

Log scales typically produce ticks at powers

d3.scaleBand() maps discrete categories to evenly spaced rectangular bands across a continuous range. It is the standard tool for bar charts. Categorical data has no numeric spacing.

```
const x = d3.scaleBand().domain(["A", "B",
↪  "C"]).range([0,300]);
```

D3 divides the range into equal bands

$$A \to 0 - 100$$
$$B \to 100 - 200$$
$$C \to 200 - 300.$$

We get the bandwidth with `bandwidth()`

```
.attr("x", d=>x(d.category)).attr("width", x.bandwidth());
```

We can add spacing between bars with

```
x.padding(p)
```

- **d3.max(), d3.min()**: Retrieves the max or min value from an iterablea. Unlike Math.max() and Math.min(), these methods ignores undefined values, which is useful for missing data.

- **d3.maxIndex(), d3.minIndex()**: Retrieves the index of the max or min value from an iterable.
- **Axes**:
  - **axis(context)**: Render the axis to the given context, which may be either a selection of SVG containers (either SVG or G elements) or a corresponding transition.
  - **d3.axisTop(scale)**: Constructs a new top-oriented axis generator for the given scale, with empty tick arguments, a tick size of 6 and padding of 3. In this orientation, ticks are drawn above the horizontal domain path.
  - **d3.axisBottom(scale)**: Constructs a new bottom-oriented axis generator for the given scale, with empty tick arguments, a tick size of 6 and padding of 3. In this orientation, ticks are drawn below the horizontal domain path.
  - **d3.axisLeft(scale)**: Constructs a new left-oriented axis generator for the given scale, with empty tick arguments, a tick size of 6 and padding of 3. In this orientation, ticks are drawn to the left of the vertical domain path.
  - **d3.axisRight(scale)**: Constructs a new right-oriented axis generator for the given scale, with empty tick arguments, a tick size of 6 and padding of 3. In this orientation, ticks are drawn to the right of the vertical domain path.
  - **axis.scale(scale?)**: If scale is specified, sets the scale and returns the axis. If scale is not specified, returns the current scale.
  - **axis.ticks(args...)**: Sets the arguments that will be passed to scale.ticks and scale.tickFormat when the axis is rendered, and returns the axis generator. The meaning of the arguments depends on the axis' scale type: most commonly, the arguments are a suggested count for the number of ticks (or a time interval for time scales), and an optional format specifier to customize how the tick values are formatted.

    This method has no effect if the scale does not implement scale.ticks, as with band and point scales. To set the tick values explicitly, use axis.tickValues. To set the tick format explicitly, use axis.tickFormat.
  - **axis.offset(offset?)**: If offset is specified, sets the offset to the specified value in pixels and returns the axis. If offset is not specified, returns the current offset which defaults to 0 on devices with a devicePixelRatio greater than 1, and 0.5px otherwise. This default offset ensures crisp edges on low-resolution devices.
  - **axis.tickSize(size?)**: If size is specified, sets the inner and outer tick size to the specified value and returns the axis. If size is not specified, returns the current inner tick size, which defaults to 6.
  - **exis.tickSizeInner(size?)**:
  - **axis.tickSizeOuter(size?)**:
  - **axis.tickFormat(format?)**: If format is specified, sets the tick format function and returns the axis. If format is not specified, returns the current format function, which defaults to null. A null format indicates that the scale's default formatter should be used, which is generated by calling scale.tickFormat. In this case, the arguments specified by axis.tickArguments are likewise passed to scale.tickFormat.

```
0  axis.tickFormat(d3.format(",.0f"));
```

Formats integers with comma-grouping for thousands

More commonly, a format specifier is passed to axis.ticks

```
axis.ticks(10, ",f")
```

- **axis.tickValues(values?)**: If a values iterable is specified, the specified values are used for ticks rather than using the scale's automatic tick generator. If values is null, clears any previously-set explicit tick values and reverts back to the scale's tick generator. If values is not specified, returns the current tick values, which defaults to null.
- **axis.tickPadding(padding?)**: If padding is specified, sets the padding to the specified value in pixels and returns the axis. If padding is not specified, returns the current padding which defaults to 3 pixels.
- **axis.tickArguments([args])**: If arguments is specified, sets the arguments that will be passed to scale.ticks and scale.tickFormat when the axis is rendered, and returns the axis generator. The meaning of the arguments depends on the axis' scale type: most commonly, the arguments are a suggested count for the number of ticks (or a time interval for time scales), and an optional format specifier to customize how the tick values are formatted.

- **The return value of axisTop() and others**: The return value is the generator function that builds the axis.

We can then call this generator passing in selectors so that an axis is built as a child of that selector.

- **Axis example**:

```
// the scale function
var xScale = d3.scaleLinear()
  .domain([0,d3.max(data)])
  .range([0,300]);

var xAxis = d3.axisTop()
xAxis.scale(xScale);
svg.append("g").call(xAxis);
```

Where

```
svg.append("g").call(xAxis);
```

means "Apply the axis generator to this <g> element, causing it to create ticks, labels, and lines inside it."

We can also apply a transform to the group element

```
let xAxis = d3.axisBottom(xScale);
svg.append("g").attr("transform",
    ↪ `translate(${margin.left},${H})`).call(xAxis);
```

And for a y-axis,

```
let yAxis = d3.axisLeft(yScale);
svg.append("g").attr("transform", `translate(${margin.left},
    ↪ 0)`).call(yAxis);
```

- **Targeting the axis text**

```
let xAxis = d3.axisBottom(xScale);
svg.append("g")
    .call(xAxis)
    .selectAll("text")
    .attr()...
```

- **Adding text**:

```
selection.text(value?)
```

If a value is specified, sets the text content to the specified value on all selected elements, replacing any existing child elements. If the value is a constant, then all elements are given the same text content; otherwise, if the value is a function, it is evaluated for each selected element, in order, being passed the current datum (d), the current index (i), and the current group (nodes), with this as the current DOM element (nodes[i]). The function's return value is then used to set each element's text content. A null value will clear the content.

If a value is not specified, returns the text content for the first (non-null) element in the selection. This is generally useful only if you know the selection contains exactly one element.

- **Inner html**:

```
selection.html(value?)
```

If a value is specified, sets the inner HTML to the specified value on all selected elements, replacing any existing child elements. If the value is a constant, then all elements are given the same inner HTML; otherwise, if the value is a function, it is evaluated for each selected element, in order, being passed the current datum (d), the current index (i), and the current group (nodes), with this as the current DOM element (nodes[i]). The function's return value is then used to set each element's inner HTML. A null value will clear the content.

If a value is not specified, returns the inner HTML for the first (non-null) element in the selection. This is generally useful only if you know the selection contains exactly one element.

- **Date object**: The JavaScript Date object is a built-in utility for working with dates and times
  - **new Date():** Creates an object for the current date and time in the local time zone.
  - **new Date(timestamp):** Creates a date from a specified number of milliseconds since the epoch.
  - **new Date(dateString):** Parses a date string (e.g., ISO 8601 format like "2023-12-25T10:00:00").
  - **new Date(year, month, day, hours, minutes, seconds, milliseconds):** Creates a date from specific components. Note that months are zero-indexed (0 for January, 11 for December).

Category	Method Examples (Local Time)	Description
Get	getFullYear(), getMonth(), getDate(), getHours(), getMinutes(), getSeconds(), getMilliseconds()	Retrieve specific components of the date.
Set	setFullYear(), setMonth(), setDate(), setHours(), setMinutes(), setSeconds(), setMilliseconds()	Modify specific components of the date.
Utility	getTime(), setTime(milliseconds), Date.now(), toISOString(), toLocaleString()	Get/set the timestamp, get current time as timestamp, format the date as a string.

- **Margins**: We create a group element inside the root svg, then apply a transform based on set margins

```
{
    const svg = d3.create("svg");
    const group = svg.append("g");

    let margin = {left: 50, right: 25, bottom: 50, top: 0}

    let W = 600, H = 400;
    svg.attr("width",
    ↪ W+margin.left+margin.right).attr("height",
    ↪ H+margin.top+margin.bottom);

    group.attr("transform", `translate(${margin.left},
    ↪ ${margin.top})`);

    return svg.node();
}
```

3.1 Geospatial data with d3 and GeoJSON

- **Intro**: D3 uses GeoJSON as the geometric description of the world, and then uses a projection to convert that spherical geographic data into 2D screen coordinates.

  The pipeline is basically:

  1. GeoJSON stores shapes in longitude and latitude.
  2. A D3 projection transforms those longitude/latitude values into x/y pixel positions.
  3. A geographic path generator turns the transformed geometry into SVG path data, or draws it directly to canvas.

- **d3.extent**: Returns the minimum and maximum value in the given iterable using natural order. If the iterable contains no comparable values, returns [undefined, undefined]. An optional accessor function may be specified, which is equivalent to calling Array.from before computing the extent.

```
d3.extent(iterable[,accessor])
```

- **d3.path**: d3.path() is a D3 utility for building vector drawing commands programmatically. you call drawing methods such as moveTo, lineTo, arc, or bezierCurveTo.

  D3 stores those commands internally, and calling .toString() gives you the SVG path data string

```
const p = d3.path();

p.moveTo(10, 10);
p.lineTo(100, 10);
p.lineTo(100, 100);
p.closePath();

console.log(p.toString());
```

- **Manual projection**: We can define two scales, one for the min and max longitude (x-axis), and one for the min and max latitude (y-axis)

```
projection0x = d3.scaleLinear().domain(d3.extent(data.coordi
↪  nates[0][0].map(d => d[0]))).range([0,w])
projection0y = d3.scaleLinear().domain(d3.extent(data.coordi
↪  nates[0][0].map(d => d[1]))).range([h,0])
```

  Then,

```
// Manual projection (don't do this)
{
  //Create SVG element
  var mapSvg = d3.create("svg")
    .attr("width", w)
      .attr("height", h);

  var path = d3.path();

  path.moveTo(projection0x(data.coordinates[0][0][0][0]),
  ↪ projection0y(data.coordinates[0][0][0][1]));
  data.coordinates[0][0].forEach(d =>
  ↪ path.lineTo(projection0x(d[0]), projection0y(d[1])))

  mapSvg.append("path")
    .attr("d", path)
    .style("stroke", "black")
    .style("fill", "none");

  return mapSvg.node();
}
```

- **Albers projection (geoAlbersUsa())**: d3.geoAlbersUsa() is a D3 map projection designed specifically for maps of the United States.

  when you use geoAlbersUsa(), D3 automatically does two things:

  - projects geographic coordinates into screen coordinates
  - repositions Alaska and Hawaii so they appear near the lower-left of the map instead of in their true geographic locations

  That is why it is so convenient for U.S. choropleths and state maps.

```
projection1 = d3.geoAlbersUsa().fitSize([w,h], data)
```

Then,

```
{
  var projection = projection1;

  //Create SVG element
  var mapSvg = d3.create("svg")
    .attr("width", w)
      .attr("height", h);

  var path = d3.geoPath()
    .projection(projection);

  mapSvg.append("path")
    .datum(data)
    .attr("d", path)
    .style("stroke", "black")
    .style("fill", "none");

  return mapSvg.node();
}
```

- **Other projections**:
  - d3.geoTransverseMercator

Observable Plot

- **Intro**: Created by the creators of D3, observable Plot follows a grammar-of-graphics style approach:
  - You describe what you want to visualize
  - The library determines how to render it
  - Plots are composed from marks applied to data

  A typical plot consists of:
  - Data
  - Encodings (x, y, color, etc.)
  - Marks (bars, dots, lines, etc.)
  - Optional transforms (grouping, binning, aggregation)

```
const plot = Plot.plot({
    marks: [
        Plot.barY(data, { x: "date", y: "count" })
    ]
});

document.body.append(plot);
```

- **Marks**: Marks define how data is rendered.
  - **Bars**:
  - **Dots / scatter**:
  - **Lines**:
  - **Areas**:
- **Automatic Aggregation**: Plot can compute statistics for you. For example, counting occurrences

```
Plot.barY(data, Plot.groupX({ y: "count" }, { x:
↪  "crash_type" }))
```

  This groups rows by crash_type and counts them.
- **Encodings**: Encodings map data fields to visual properties. Common channels are
  - x — horizontal position
  - y — vertical position
  - color — categorical or quantitative color
  - fill / stroke — appearance
  - size — point size
  - opacity
- **Faceting and Grouping**: You can split data into panels or aggregate within categories.

```
Plot.barY(
    data,
    Plot.groupX({ y: "count" }, { x: "date" })
)
```

Notice that we use *count* above, but we also have options

  - **count** - the number of observations in each group
  - **sum** - the sum of values
  - **proportion** - like sum, but divided by the total
  - **proportion-facet** - like sum, but divided by the facet's total
  - **min** - the minimum value
  - **max** - the maximum value
  - **mean** - the mean (average) value
  - **median** - the median value
  - **mode** - the value with the most occurrences
  - **variance** - an unbiased estimator of population variance
  - **deviation** - the standard deviation
  - **first** - the first value (in input order)
  - **last** - the last value (in input order)

- **Axes and Scales**: Plot automatically creates axes based on data types.

```
Plot.plot({
    x: { label: "Date" },
    y: { label: "Number of crashes" },
    marks: [...]
})
```

GeoJSON

- **Intro**: GeoJSON is a standard text format for representing geographic data using JSON.

  It is used to describe things like:

  - a single point, such as a store or landmark
  - a line, such as a road or route
  - an area, such as a city boundary or lake
  - collections of those objects, along with extra data about them

  A simple example of a point looks like this:

```
{
    "type": "Point",
    "coordinates": [-88.7504, 41.9295]
}
```

  the order is usually [longitude, latitude], not latitude first

- **Geometry types**: GeoJSON supports several geometry types:

  - **Point**: One location.
  - **MultiPoint**: Several separate locations.
  - **LineString**: A sequence of points connected by straight segments.
  - **MultiLineString**: Several lines.
  - **Polygon**: A closed shape, such as a parcel or region.
  - **MultiPolygon**: Several polygons.
  - **GeometryCollection**: A group of different geometries.

- **Features**: Very often, GeoJSON is not just raw geometry. It is wrapped in a Feature, which adds metadata:

```
{
    "type": "Feature",
    "geometry": {
        "type": "Point",
        "coordinates": [-88.7504, 41.9295]
    },
    "properties": {
        "name": "DeKalb",
        "state": "Illinois"
    }
}
```